

Facial Expression Recognition Framework

Background and Motivation - I selected this topic because emotion recognition is widely used in smart applications, customer analysis, and human-computer interaction systems. I wanted to understand how CNN models can identify facial expressions and emotions from images. Facial expression recognition is a key area in computer vision and artificial intelligence. It is widely used in human-computer interaction, surveillance systems, and emotion-aware applications. This project aims to build a deep learning model that can automatically detect human emotions from facial images.

Problem Statement - The objective of this project is to develop a CNN-based model that classifies facial images into seven emotion categories: angry, disgust, fear, happy, neutral, sad, and surprise.

Selected Dataset and Dataset Link

Dataset: FER2013 (Facial Expression Recognition 2013)

Link: <https://www.kaggle.com/datasets/msambare/fer2013>

Dataset Description - The FER2013 dataset contains grayscale facial images of size 48x48 pixels. It includes images labeled into seven emotion categories. The dataset is split into training and testing sets.

Number of Images

Training images: ~28,709

Testing images: ~7,178

Total images: ~35,887

Number of Classes - 7 classes: Angry, Disgust, Fear, Happy, Neutral, Sad, Surprise

Image Type - Grayscale facial expression images (48x48 pixels)

Class Distribution - Dataset is slightly imbalanced, with fewer samples for 'disgust' compared to other classes such as 'happy' and 'neutral'.

Research Questions

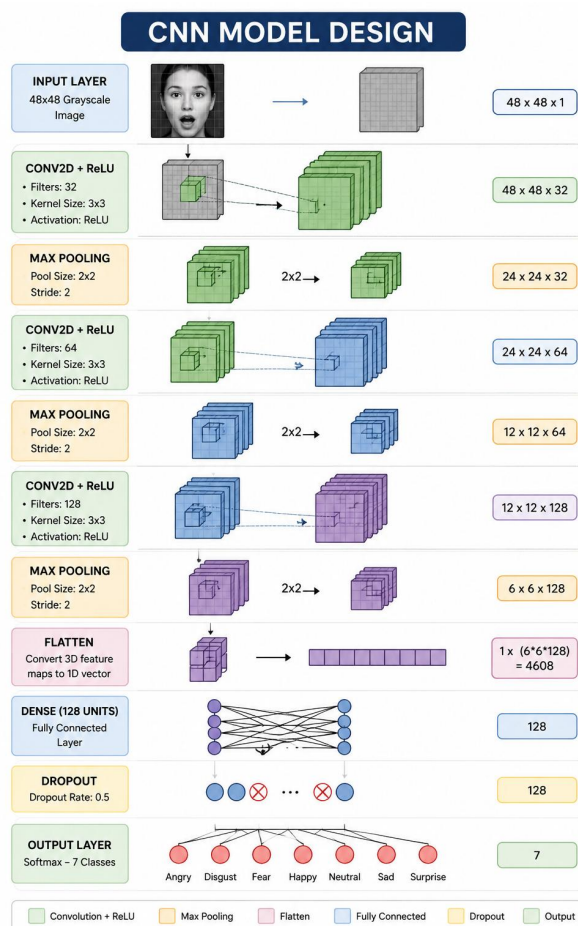
1. Can CNN effectively classify facial expressions from grayscale images?
2. Does transfer learning improve accuracy compared to a custom CNN?
3. Which emotion classes are most difficult to classify?
4. How does data augmentation affect model performance?

Expected Results - The CNN model is expected to achieve moderate accuracy, while transfer learning models like MobileNetV2 or ResNet50 are expected to outperform the custom CNN.

Proposed Methodology

1. Data preprocessing and normalization
2. Data augmentation
3. CNN model development
4. Model training and validation
5. Transfer learning implementation
6. Model evaluation and comparison

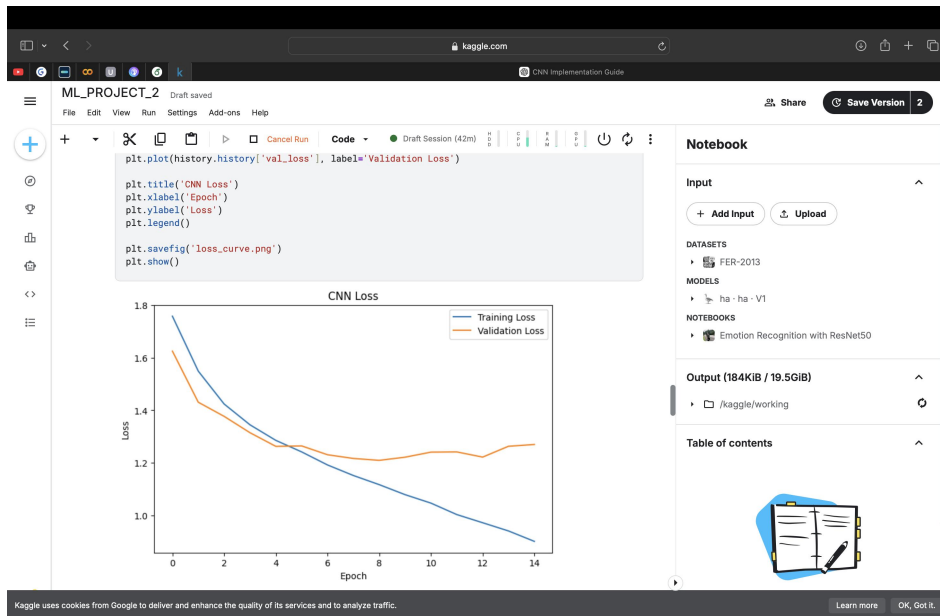
CNN Model Design



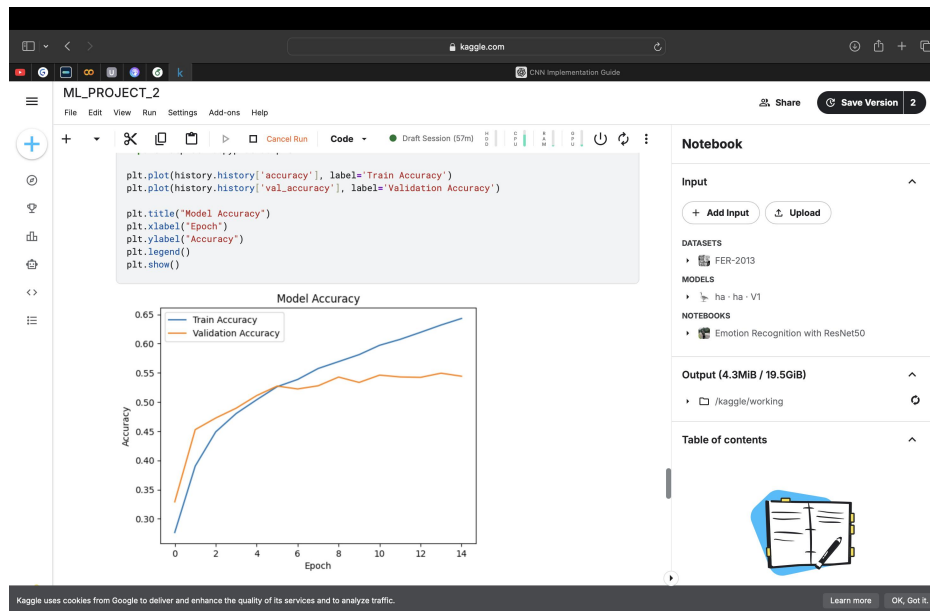
Transfer Learning Models Used - MobileNetV2 and ResNet50 pretrained on ImageNet dataset are used for comparison with CNN model.

Evaluation Metrics - Accuracy, Loss, Precision, Recall, F1-score, Confusion Matrix

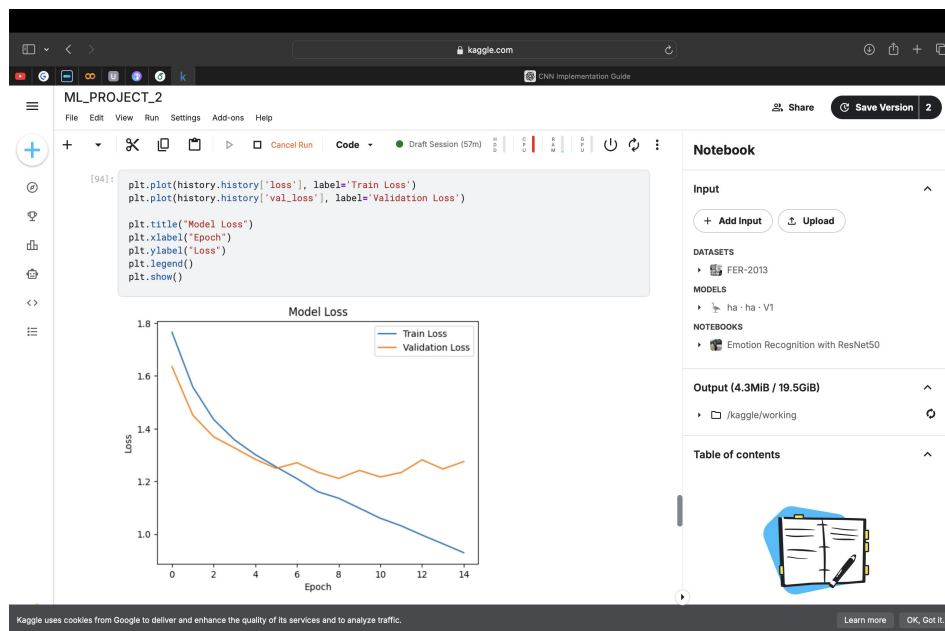
Expected Figures and Tables



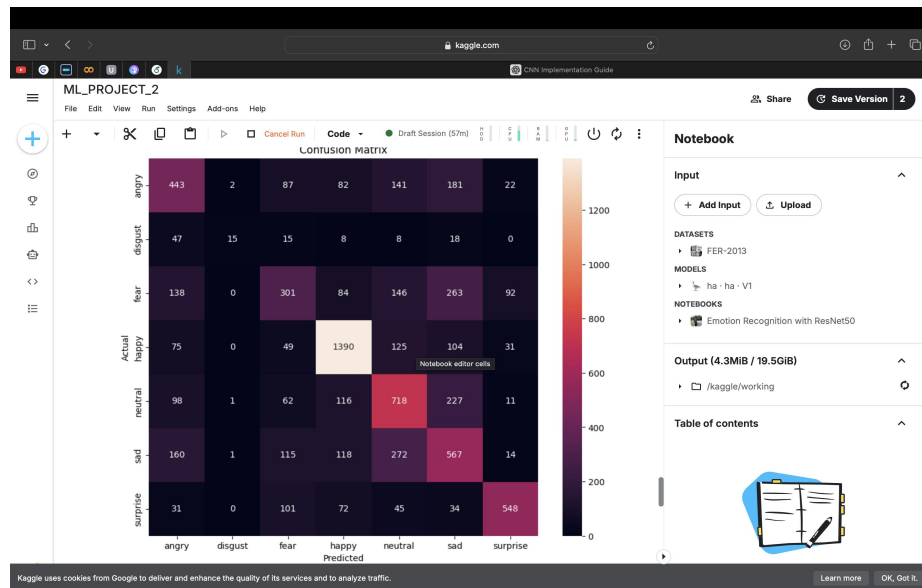
CNN LOSS



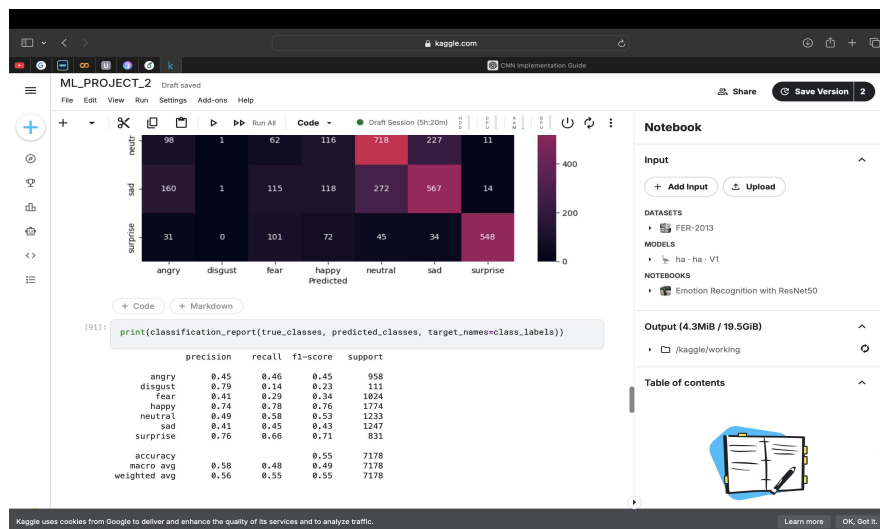
MODEL ACCURACY



MODEL LOSS



CONFUSION MATRIX



Classification Report

ML_PROJECT_2 Draft saved

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Code Draft Session (5h:23m)

```
[108]: print('cnn_model' in globals())
True

[113]: print(cnn_model.input_shape)
(None, 48, 48, 1)

[114]: cnn_model.evaluate(test_generator)
225/225 14s 61ms/step - accuracy: 0.5548 - loss: 1.2469
[11. [1.246938943862915, 0.5547506213188171]

cnn_model.evaluate(test_generator)
225/225 7s 32ms/step - accuracy: 0.5548 - loss: 1.2469
[11. [1.246938943862915, 0.5547506213188171]
```

+ Code + Markdown

Notebook editor cells

Notebook

Input

+ Add Input Upload

DATASETS

FER-2013

MODELS

ha · ha · V1

NOTEBOOKS

Emotion Recognition with ResNet50

Output (4.3MiB / 19.5GiB)

/kaggle/working

Table of contents

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CNN Accuracy Result